Bowen Yang

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Research Interests

Robot learning and embodied AI, with a focus on vision–language–action models, structured representations for robotic manipulation, and robust autonomous systems.

Education

Shanghai Jiao Tong University Aug 2023 –
Bachelor's studies in French; Minor in Information Engineering Jun 2027 (expected)
SJTU Paris Elite Institute of Technology. Advisor: **Prof. Junchi Yan**.
Average score: **91.69** · Rank: **2/77**

Publications

ELAN4D: Embodiment-Centric 4D Supervision for Vision-Language-Action Models via Plug-and-Play Adaptation 2026
Zeyuan He*, **Bowen Yang***, Zhirui Fang*, Keru Zhou*, Lei Jiang, Jingjing Qian, Fan Mo, Junchi Yan, Philip Torr, Xiu Li, Li Jiang, Jialin Yu†
Conference on Robot Learning (CoRL)

GuidedVLA: Specifying Task-Relevant Factors via Plug-and-Play Action Attention Specialization 2026
Xiaosong Jia*, **Bowen Yang***, Zuhao Ge*, Xian Nie*, Yuchen Zhou*, Cunxin Fan*, Yufeng Li, Yilin Chai, Chao Jing, Zijian Liang, Qingwen Bu, Haidong Cao, Chao Wu, Qifeng Li, Zhenjie Yang, Chenhe Zhang, Hongyang Li, Zuxuan Wu†, Junchi Yan†, Yu-Gang Jiang†
Robotics: Science and Systems (RSS)

FOMI: A Force-Feedback Manipulation Interface for Teaching Contact-Rich Manipulation 2026
Yishu Shen, Yiming Luo, Ruilin Wang, **Bowen Yang**, Shuai Yang, Fangyi Xu, Yujun Shen, Fu Zhang, Boyu Zhou, Yinghao Xu†
arXiv preprint arXiv:2610.19839
* Equal Contribution; † Corresponding Author

Research Experience

Contact-Rich Manipulation: Demonstration Collection and Policy Learning June 2026 – present
Algorithm Engineer Intern, **RobbyAnt** · Co-author · FOMI, arXiv 2026

- Built **80 Hz synchronized** camera, tracker, and encoder acquisition with a **data-check tool**, and extended **tracker-to-robot calibration** (forward kinematics, nonlinear least squares).
- Implemented **relative-pose actions** and **real-time action chunking** for $\boldsymbol{\pi_{0.5}}$ **fine-tuning**.
- Deployed and evaluated policies on **Franka and Flexiv**: **78.4%** average success on four contact-rich tasks (100 demonstrations each), and **68.2%** after transfer to a dual-arm Flexiv.

Embodiment-Centric 4D Supervision for Robot Learning Mar 2026 – May 2026
Co-first author · ELAN4D, **CoRL 2026**
Research collaboration with Torr Vision Group, University of Oxford.

- Proposed using **URDF-based forward kinematics** to derive future **robot-joint and end-effector keypoint tracks** from proprioception as low-cost **4D supervision**.
- Proposed a **ControlNet-style residual branch** that **isolates auxiliary gradients** from the pretrained VLM and keeps the base policy's inference interface unchanged.
- Co-designed **ablations** on 4D supervision, prediction location, and **robot-keypoint versus whole-scene tracking** to isolate the effect of each component.
- ELAN4D raised LIBERO-Plus success from **53.6% to 67.6%** (π_0) and **73.6% to 78.2%** ($\pi_{0.5}$).

Task-Relevant Representation Learning for VLA Policies Sept 2025 – Jan 2026
Co-first author · RA, RethinkLab, SJTU · GuidedVLA, **RSS 2026** · Project
Advisors: Prof. Junchi Yan and Prof. Xiaosong Jia.

- Led **codebase implementation and simulation experiments** for GuidedVLA, which explicitly supervises task-relevant factors in a vision–language–action policy.
- Implemented **specialized action-attention heads** for **object grounding**, **temporal skill recognition**, and **depth-based geometric perception** in the π_0 action decoder.
- Integrated a **ControlNet-style residual adapter** for plug-and-play factor supervision.
- Evaluated GuidedVLA against the baseline: average success rose from **68.2% to 75.4%** on LIBERO-Plus and from **77.4% to 90.6%** on RoboTwin 2.0.
- Optimized the openpi PyTorch training pipeline with **mixed precision** and **torch.compile**.

Selected Projects

RoboMaster University Championship — Team Jiao Loong Sep 2023–Aug 2025
Vice-Captain & Algorithm Lead (Aug 2024–Aug 2025)

- Led a **15-member algorithm team** to **national championships in 2024 and 2025**, and led **recruitment and training of 60+ new members** and a **32-team on-campus competition**.
- Developed **FAST-LIO2 LiDAR–inertial SLAM** with **GICP map registration** and **PTP-synchronized LiDAR timing** for localization during fast robot motion.
- Built a motion planner combining **kinodynamic A$_*$** and **model predictive control (MPC)** for navigation and trajectory tracking on embedded hardware.
- Deployed **TensorRT-accelerated YOLO** on **Jetson Orin** with **EKF-based motion prediction** to track opponent robots through visual occlusion.
- Won the **2025 RM Award (Algorithm)** for **sentry SLAM and navigation**, one of three algorithm recipients selected through review and interview.

Honors & Awards

National Scholarship 2024, 2025
Ministry of Education of China

National Champion, RoboMaster University Championship 2024, 2025
Team Jiao Loong, Shanghai Jiao Tong University

RM Award (Algorithm), RoboMaster University Championship 2025
Individual technical honor; one of three algorithm recipients in 2025

Finalist, Mathematical Contest in Modeling (MCM) 2025
Problem B: environmental dynamic system evolution and multi-objective optimization modeling

Bronze Medal, National Olympiad in Informatics (NOI) Winter Camp 2021
China Computer Federation (CCF)

Platinum Division, USACO 2020
USA Computing Olympiad (USACO)

Technical Skills

Programming: Python, C, C++
Machine Learning & Vision: PyTorch, TensorFlow, OpenCV, TensorRT
Robotics & Development: ROS/ROS 2, Docker, Git, LaTeX

Languages

English: TOEFL iBT **5.5/6** (ETS comparable total: **110/120**), 8 Mar 2026.
French: DELF B2 (CEFR B2), diploma issued 26 Apr 2026.